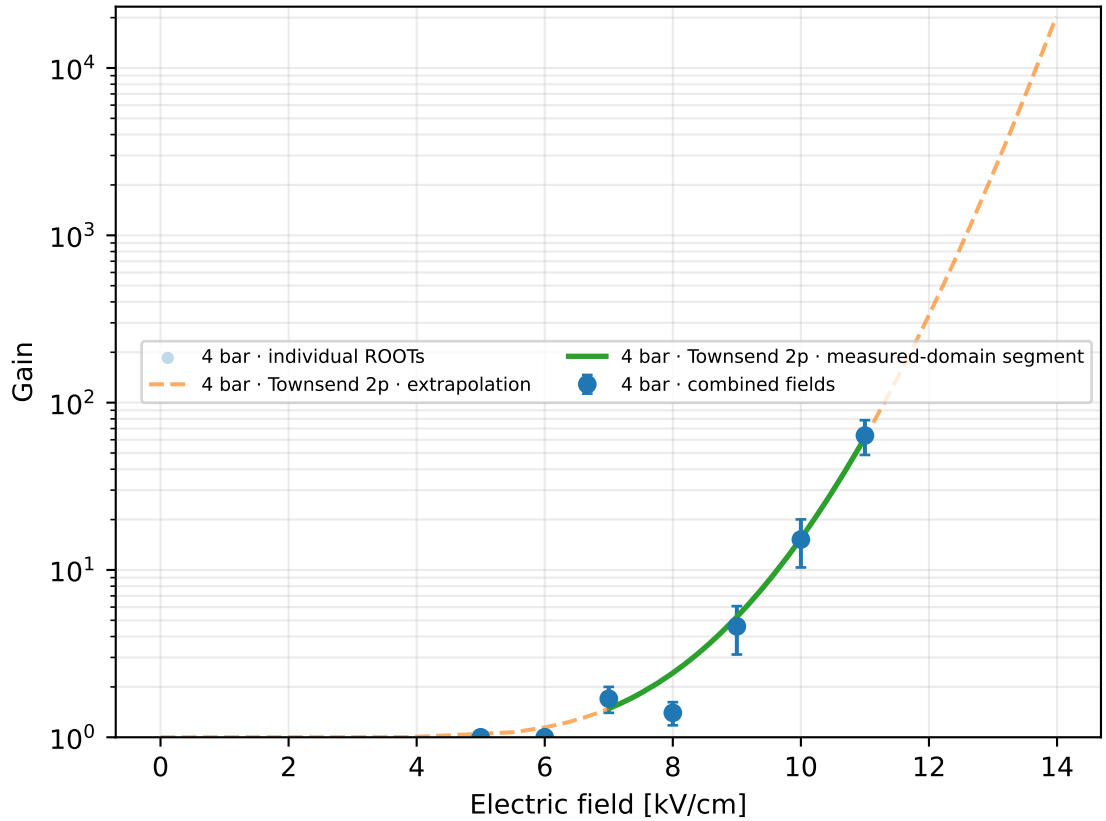
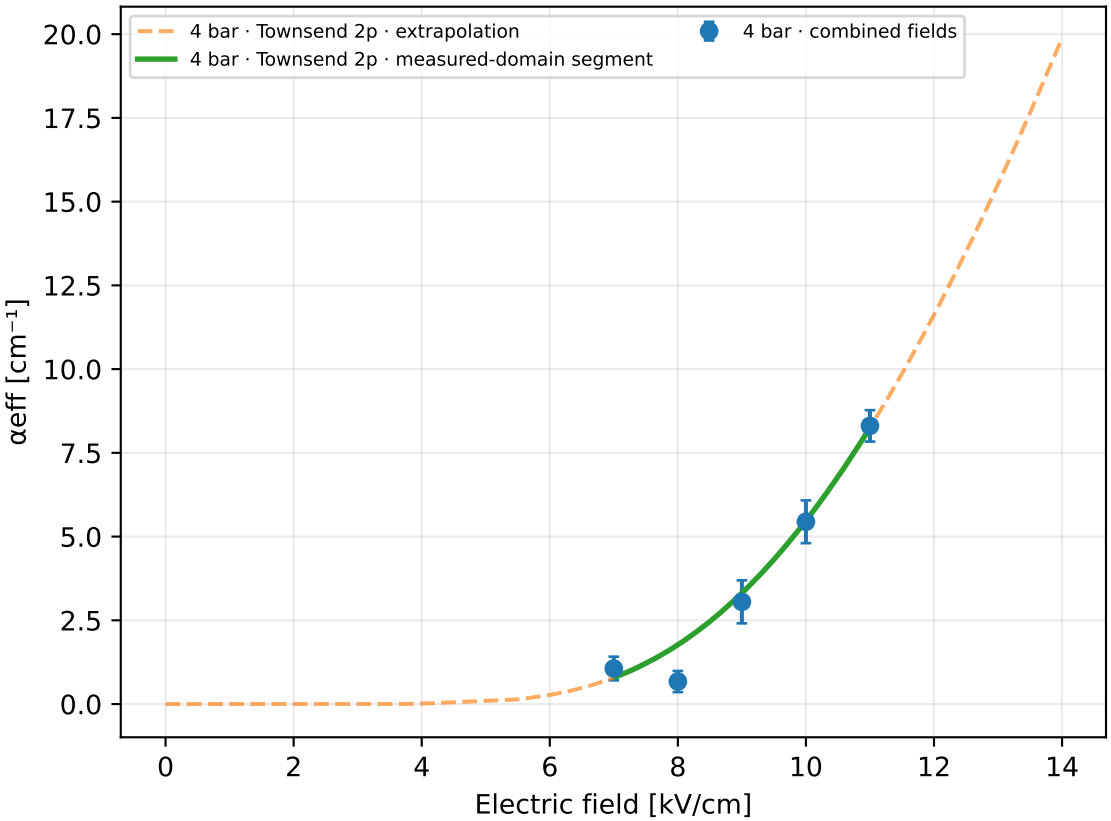
ArXe · Ar 99 % / Xe 1 % · gap 5 mm

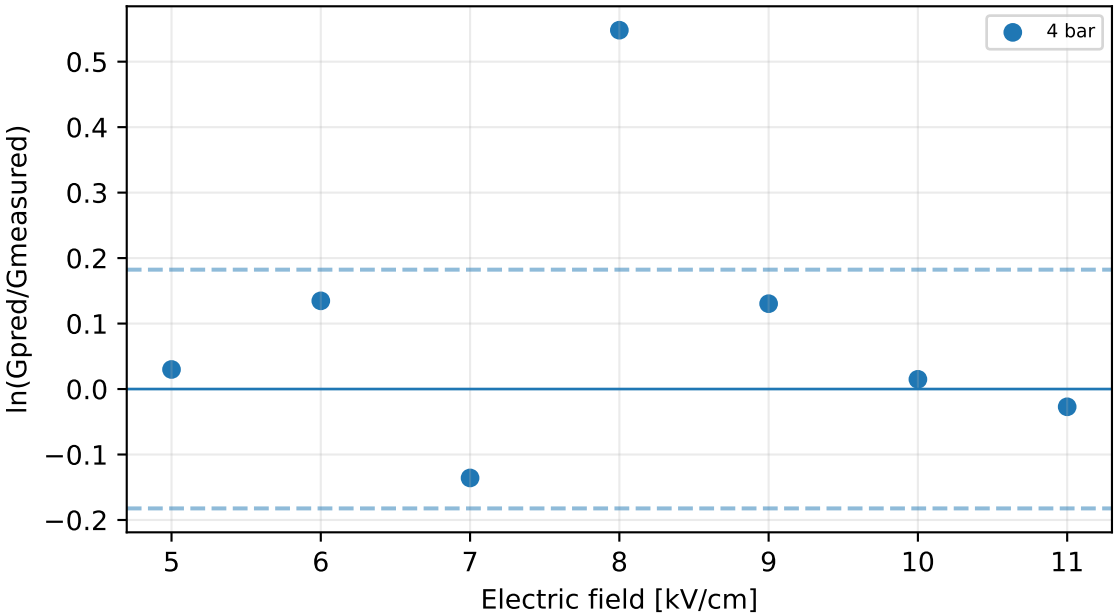
Gain vs electric field



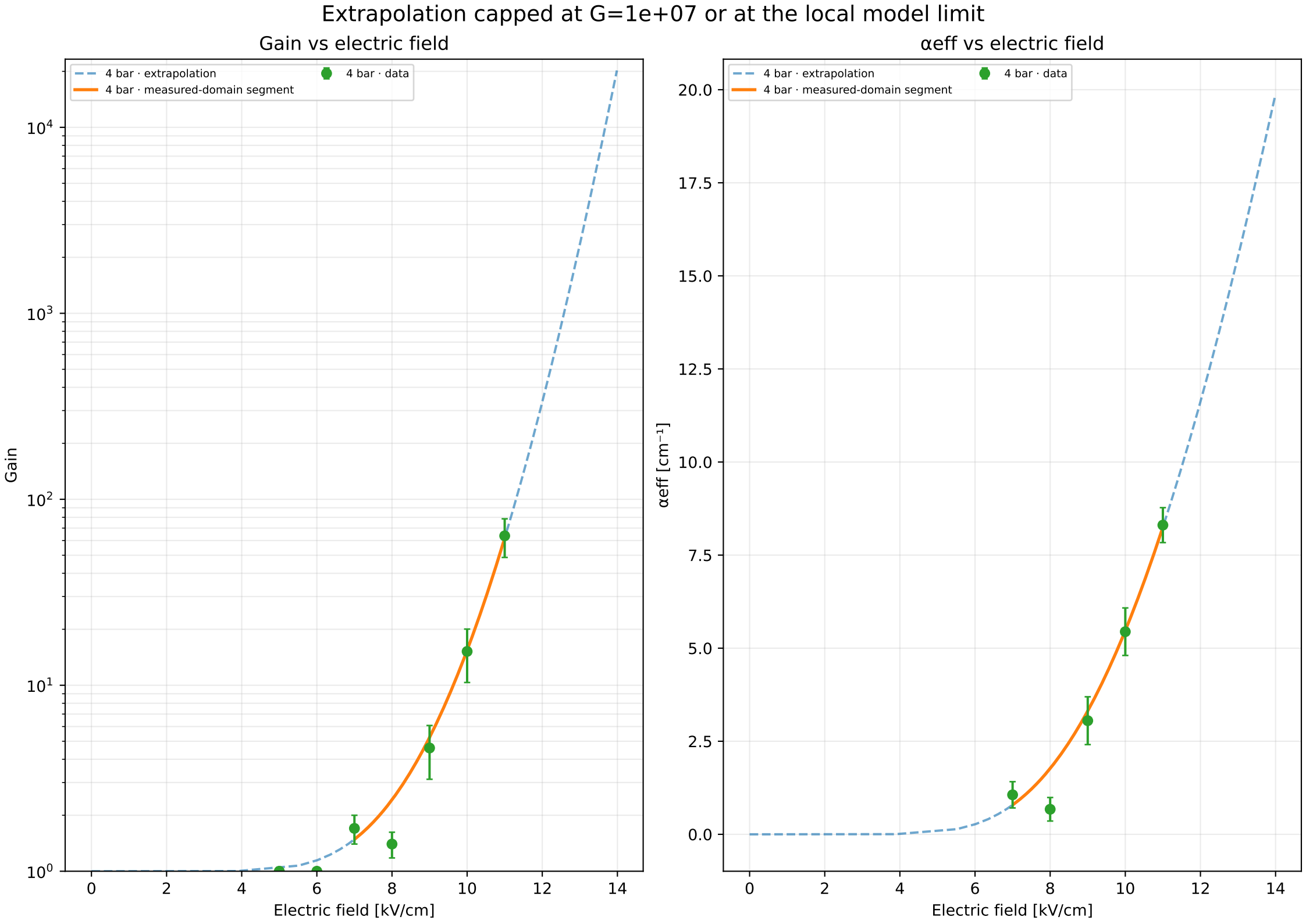
Effective Townsend coefficient vs electric field



Gain-space residuals (±ln 1.2 guides)




```
ROOT runs: 7
Unique physical points: 7
Pressures: 4 bar
Selected model: Townsend 2p
Status: rejected_large_training_residuals
Training RMSE: ×1.574
Cross-validation median: ×1.307
Cross-validation worst: ×1.875
Reduced-field range: 1.75–2.75 kV cm⁻¹ bar⁻¹
A=125.26, B=11.292, m=0, n=1
Covariance condition: 3.55e+03
Warnings: large_training_residuals
4 bar display limit: G=2.02e+04 (local inversion limit)
```


Extrapolation capped at G=1e+07 or at the local model limit
Gain vs electric field
4 bar · extrapolation
4 bar · measured-domain segment
4 bar · data
Gain
Electric field [kV/cm]
αeff vs electric field
αeff [cm⁻¹]
Electric field [kV/cm]

Exact combined points used by the fit · page 1/1

Repeated ROOTs at the same field are pooled once; all source ROOTs remain on disk.

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm⁻¹]	npe	runs
4	5	1.25	1	0	—	10	1
4	6	1.5	1	0	—	10	1
4	7	1.75	1.7	0.3	1.06126	10	1
4	8	2	1.4	0.221	0.672944	10	1
4	9	2.25	4.6	1.48	3.05211	10	1
4	10	2.5	15.2	4.85	5.44259	10	1
4	11	2.75	63.6	14.9	8.30523	10	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	2.09	×1.31	×1.88	no	large training residuals